

Leveraging Large Language Models for Dynamic Multi-Objective Optimization in UAV Sensor-Target Assignment

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Abstract: The multi-sensor target assignment problem in Unmanned Aerial Vehicles (UAVs) plays a vital role in optimizing a range of tasks, including navigation, target tracking, and obstacle avoidance. It focuses on the efficient allocation of sensors to specific targets, improving task performance, accuracy, and operational efficiency. Traditional multi-objective optimization algorithms often rely on static operators, which limits their adaptability to dynamic environments. To address this challenge, we introduce LLM-MOEA, a novel Large Language Model (LLM)-driven multi-objective evolutionary algorithm. LLM-MOEA dynamically adjusts optimization operators, leveraging population-level insights through an innovative prompt design and integrates the optimization trajectory to enhance the algorithm's decision-making capabilities. Experimental results demonstrate that LLM-MOEA significantly outperforms traditional algorithms in terms of solution quality, convergence speed, and robustness. Ablation studies further confirm the importance of key components. This work highlights the potential of combining LLMs with evolutionary algorithms to solve complex multi-objective optimization problems and enhance the flexibility of optimization methods.

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Keywords: Sensor-target assignment, Adaptive operator selection, Large language models, Multi-objective evolutionary algorithm.

1. INTRODUCTION

Advanced electronics and artificial intelligence have transformed modern unmanned aerial Vehicles (UAVs) into cognitive platforms that integrate diverse sensor suites—such as electro-optical, radar, and micro electronic support measures. To harness the full potential of these multi-sensor systems, we need to develop systematic allocation strategies that operate within practical constraints and balance cost-effectiveness with resource optimization. Essentially, this challenge stems from the Weapon Target Assignment (WTA) problem (Kline et al., 2019), and its research trajectory mirrors that of WTA, evolving from exact algorithms to heuristic approaches.

In early research stages, scholars predominantly employed exact algorithms to address the Sensor-Target Assignment (STA) problem. Nash (1977) pioneered the application of optimization techniques in this domain, implementing linear programming (LP) for deterministic allocation design between sensors and tracked targets. Subsequently, Castanon (1997) extended this approach by developing multi-objective dynamic programming (DP) for as-

signment schemes. Parallel research efforts systematically investigated the implementation of LP (Schumacher et al., 2004), DP (Cai et al., 2006), and branch-and-bound methods (Ahuja et al., 2007) in resource allocation contexts. Given the inherent NP-complete nature of STA problems, where computational complexity escalates exponentially with problem scale, researchers progressively shifted focus toward computationally efficient heuristic approaches for practical implementations. Wang et al. (2013) developed a collaborative memetic particle swarm optimization (PSO) framework integrating neighborhood search mechanisms to solve STA problems in air defense systems. Building upon this foundation, Xu et al. (2016) enhanced solution quality through a self-adaptive quantum PSO algorithm with memetic collaboration. Another innovation by Jian and Chen (2015) combined STA with WTA problems, resolved through modified genetic algorithms (GA). The research community has further expanded heuristic methodologies to various allocation problems, implementing meta-heuristic techniques including ant colony optimization (Hu et al., 2018), artificial bee colony (Chang et al., 2018), and whale optimization algorithm (Zhang et al., 2023).

The application of heuristic algorithms often encounters performance dependencies on operator selection and parameter configuration. Consequently, substantial research efforts have been devoted to developing systematic methods for controlling parameters and operators in heuristic frameworks to enhance optimization efficacy. This control mechanism inherently aligns with the Markov Decision Process (MDP) framework in reinforcement learning (RL), motivating growing interest in integrating RL with evolutionary algorithms (EA) to establish advanced optimization paradigms. For instance, Wang et al. (2023) implemented Deep Q-Network (DQN) in operator selection for multi-objective artificial bee colony algorithms to address WTA problems in unmanned vehicles, while Chen et al. (2020) investigated parameter control in GAs through hybridized SARSA and Q-learning techniques. Nevertheless, it is noteworthy that RL-based approaches typically require training data and time investments, which impose practical constraints on their real-world deployment.

The recent advancement of Large Language Models (LLMs) has driven their rapid deployment across engineering, finance, biology, and other disciplines. Notably, Yang et al. (2023) pioneered the integration of LLMs into optimization science through the Optimization by PROMpting (OPRO) framework, which leverages LLMs' in-context learning capabilities by feeding optimization trajectories as input to generate optimized outputs. This approach demonstrated superior performance to manually designed prompts on the GSM8K benchmark. Subsequently, Guo et al. (2023) extended this paradigm by utilizing chain-of-thought (CoT) prompting strategies to implement four iterative optimization methods—gradient descent, hill climbing, grid search, and black-box optimization—through LLM interactions. Huang et al. (2024b) further adapted these techniques for multimodal LLMs, achieving great results on capacitated vehicle routing problems through explicit graphical representation of problem structures.

Critical analysis reveals that LLMs employed as standalone optimizers show limited advantages over conventional heuristics, prompting hybrid approaches that synergize LLMs with classical optimization (Huang et al., 2024a). Brahmachary et al. (2025) developed the Language-model-based Evolutionary Optimizer (LEO), which partitions populations into exploratory and exploitative subgroups under an EA framework. Liu et al. (2024b) innovatively deployed LLMs as adaptive crossover and mutation operators in Traveling Salesman Problem (TSP) optimization, outperforming the original OPRO framework. The research community has also explored the use of LLMs in generating evolutionary algorithms, leading to numerous innovative frameworks such as FunSearch (Romera-Paredes et al., 2024), EoH (Liu et al., 2024a), and ReEvo (Ye et al., 2024).

The current research landscape lacks investigations into applying LLMs to operator selection in multi-objective optimization algorithms. To address this gap, we introduce LLMs into the operator selection process of Non-dominated Sorting Genetic Algorithm II (NSGA-II) to enhance solution quality for STA problems. Specifically, we first construct appropriate state features for the STA problem, provide short-term optimization trajectories to

the LLM, and then require the LLM to select suitable operators from a predefined operator pool, with continuous updates during the interaction process. Experimental results demonstrate that using LLMs for operator selection significantly improves solution quality, offering a novel approach for operator selection in multi-objective optimization algorithms.

This study makes three principal contributions: 1) proposing Large Language Model-driven Multi-Objective Evolutionary Algorithm (LLM-MOEA), a novel framework for operator selection in NSGA-II to address STA problems; 2) constructing an interpretable decision-making architecture that integrates domain-specific optimization features with real-time optimization context, enabling LLMs to mimic expert-level operator selection strategies while providing human-interpretable rationale; 3) empirically validating LLM-MOEA's solution quality enhancement through experiments and demonstrating its practical applicability in real-world optimization scenarios. The paper is organized as follows: Section 2 establishes the mathematical formulation of STA; Section 3 details the design and implementation of the LLM-MOEA framework; Section 4 elaborates experimental configurations, comparative results, and ablation studies; finally, Section 5 concludes with research summary and future directions.

2. PROBLEM FORMULATION

In the context of UAV sensor-target assignment problems, well-designed allocation schemes can enhance the detection performance of multi-sensor systems while balancing operational costs. In this study, we employ information gain as a metric to quantify detection performance, which optimizes the efficiency of target trajectory information acquisition through sensor allocation. Furthermore, we integrate target prioritization and sensor collaboration capabilities to formulate the STA objective function, thereby achieving more precise and resource-efficient sensor allocation strategies. The mathematical symbols involved in the task are detailed in Table 1, and the schematic diagram of UAV sensor-target assignment is shown in Fig. 1.

Table 1. Symbol definitions for UAV sensor-target assignment problems.

Symbol	Definition
$S = \{s_i \mid i = 1, 2, \dots, M\}$	The set of sensors
$T = \{t_j \mid j = 1, 2, \dots, N\}$	The set of targets
$v_i \ (i = 1, 2, \dots, M)$	The cost of the sensor s_i
$C_j \ (j = 1, 2, \dots, N)$	The priority score of the target t_j
$T_i \ (i = 1, 2, \dots, M)$	The collaborative coefficient of the sensor s_i
$R_i \ (i = 1, 2, \dots, M)$	The capacity number of the sensor s_i
x_{ij}	The decision variable of the STA problem
G_{ij}	The information gain of the pair (s_i, t_j)

2.1 Sensor-Target Assignment Problem

In UAV-based STA scenarios, we consider a system where M sensors mounted on UAVs are tasked with tracking N targets. To address overlapping tracking scenarios where multiple sensors concurrently monitor the same target, we implement a virtual composite sensor mechanism — aggregating sensor clusters into unified logical units — ensuring one-to-one target tracking assignments. Furthermore, collaborative sensor groups are pre-calibrated based on

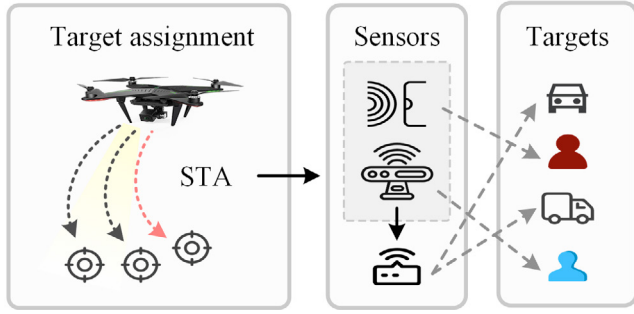


Fig. 1. Schematic diagram of sensor-target assignment.

hardware characteristics including spectral diversity and modality complementarity, with collaborative coefficients T_i formally defined to quantify performance enhancement from multi-sensor cooperative observation.

Target priority serves as a critical driver in sensor allocation strategies, reflecting a target's comprehensive threat level and tracking urgency. This metric is computed by synthesizing battlefield factors, including identity recognition, motion characteristics, and information uncertainty. We employ a normalized priority score $C_j \in [0, 1]$ where higher values indicate elevated threat levels. Information gain quantifies the observational efficacy of sensors on target states, calculated through the variation rate of covariance matrices in Kalman filter predictions to characterize uncertainty reduction in target tracking. For sensor-target pairs (s_i, t_j) , a higher information gain value G_{ij} indicates greater improvement in tracking precision. To enable efficient allocation, our framework integrates target priority C_j , information gain G_{ij} , and sensor collaborative coefficients T_i to holistically evaluate allocation quality.

2.2 Optimization Model Formulation

Capitalizing on the features of sensor-target assignment problems, we develop a bi-objective optimization model to balance information acquisition efficiency against resource consumption costs. The decision matrix is defined as $X = [x_{ij}]_{M \times N}$ where $x_{ij} = 1$ indicates assigning target t_j to sensor s_i , while $x_{ij} = 0$ signifies no assignment. The dual optimization objectives jointly address comprehensive effectiveness maximization and resource cost minimization:

$$\max F_1 = \sum_{i=1}^M \sum_{j=1}^N T_i C_j G_{ij} x_{ij}, \quad (1)$$

$$\min F_2 = \sum_{i=1}^M \sum_{j=1}^N v_i x_{ij}, \quad (2)$$

where F_1 represents the maximization of tracking capability, while F_2 denotes the minimization of resource consumption. Here, v_i corresponds to the usage cost of each sensor. The model constraints are formulated as follows:

$$\text{s.t.} \quad \begin{cases} \sum_{j=1}^N x_{ij} \leq R_i, & \forall i \in \{1, 2, \dots, M\} \\ \sum_{i=1}^M x_{ij} = 1, & \forall j \in \{1, 2, \dots, N\} \\ x_{ij} \in \{0, 1\}, \end{cases} \quad (3)$$

where the first constraint enforces that each sensor s_i must track targets within its capacity limits; the second constraint ensures each target t_j is exclusively assigned to a single sensor; the third constraint specifies variable type restrictions on the decision matrix X , requiring all elements x_{ij} to be binary variables.

3. PROPOSED METHOD

Current optimization methods using LLMs predominantly feed optimization trajectories to LLMs. This approach underutilizes critical optimization features and tends to accumulate redundant information during advanced optimization phases. In contrast, RL-based operator selection methods effectively exploit optimization features but require intensive training. To resolve these dual limitations, we propose LLM-MOE, which innovatively harnesses LLMs' reasoning capacity to systematically process optimization features in multi-objective scenarios. The framework architecture is visualized in Fig. 2.

3.1 Algorithm Framework

In this work, we propose a novel LLM-driven MOEA framework to adaptively select operators in EAs. The detailed implementation is presented in Algorithm 1. The algorithm begins by initializing key parameters, including the number of iterations T , population size N , initial operators s_0 , and the operator set $S = \{s_1, \dots, s_K\}$. A population is first initialized and evaluated based on its objective functions. The individuals are then assigned ranks through Pareto sorting to establish their relative dominance relationships within the multi-objective optimization framework (lines 1-3).

Algorithm 1 Large Language Model-driven Multi-Objective Evolutionary Algorithm

Require: Number of iterations T , population size N , set of operators $S = \{s_1, \dots, s_K\}$, initial operators s_0 , operator update interval p .

- 1: Initialize population $\mathbb{P}$.
 - 2: Evaluate objective function values.
 - 3: Assign ranks based on Pareto sorting.
 - 4: **for** every p iterations $t \leftarrow 1, 1 + p, \dots, T$ **do**
 - 5: **for** each iteration $i \leftarrow 1, \dots, p$ **do**
 - 6: Select parents using the current selection method.
 - 7: Perform a step of optimization using operator s .
 - 8: Merge parent and offspring populations.
 - 9: Perform survivor selection via Pareto sorting.
 - 10: Record the optimization trajectory.
 - 11: **end for**
 - 12: Calculate the required optimization features.
 - 13: Retrieve the optimization trajectory.
 - 14: Integrate logs to construct a comprehensive prompt.
 - 15: Select new operators based on the LLM's responses.
 - 16: **end for**
 - 17: **return** final population $\mathbb{P}^*$.
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During the iterative process, parent individuals are first selected using the current selection strategy, and an optimization step is executed by applying the recommended operators to generate offspring (lines 6-7). Then, the parent and offspring populations are merged, after which

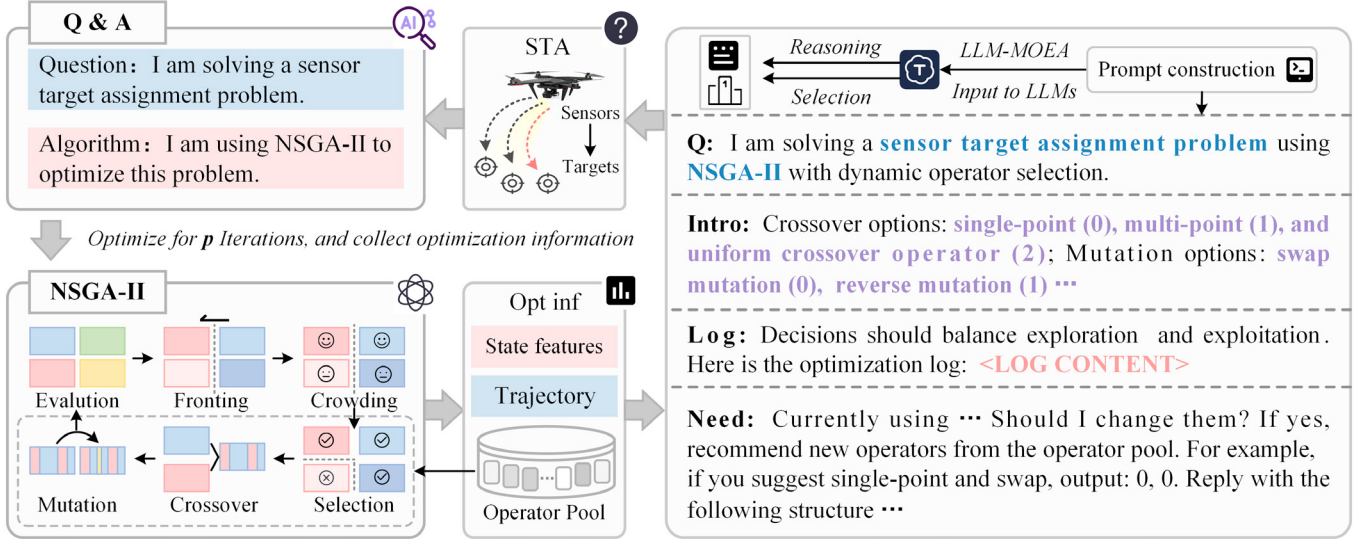


Fig. 2. Illustration of the LLM-MOEA framework. The problem type and employed algorithm are first determined. Subsequently, p iterations of the multi-objective optimization algorithm are performed, during which the optimization trajectory and corresponding state features are collected. A comprehensive prompt is then constructed, where the blue text indicates the utilized method and the problem to be addressed, the purple segment represents the operator pool, and the red segment denotes the specific optimization logs. This prompt is then fed into the LLM, which is tasked with providing the rationale and selection outcomes. Finally, the returned results are integrated back into the optimization algorithm to guide the subsequent p iterations of optimization.

survival selection is performed based on Pareto sorting to retain high-quality solutions (lines 8-9). Upon completing p iterations, the algorithm calculates optimization features such as the optimization progress and the best assignment approach. These features and the optimization trajectory in p generations are integrated into a structured natural-language prompt that contextualizes the current optimization state. The prompt is then processed by the large language model, which analyzes historical performance and recommends the most suitable operators for subsequent iterations (lines 12-15). The selected operators are dynamically incorporated into the evolutionary process to enhance adaptability.

3.2 Prompt Design

The in-context learning ability of large language models provides fundamental support for their role in assisted optimization tasks. In classical frameworks such as OPRO (Yang et al., 2023) and LMEA (Liu et al., 2024b), prompt design relies solely on optimization trajectories; however, this approach lacks key decision-support information for adaptive operator selection (AOS) tasks. Whether the AOS architecture is stateless or state-based, the incorporation of optimization features is essential. For example, the Compass (Maturana and Saubion, 2008) method plays a significant role in balancing diversity with fitness variations, while DE-DDQN (Sharma et al., 2019) constructs a 99-dimensional state space using 18 distinct features.

Therefore, in this study, we integrate both optimization trajectories and optimization features as combined optimization logs input to the LLM. Specifically, for the optimization trajectory, we feed the trajectory data from the first p iterations into the LLM in a single pass. Each iteration requires recording: 1) the fitness values of newly added non-dominated individuals in the population before

Table 2. Optimization features for LLM-MOEA and their definitions.

Feature	Description
f_1	The average fitness of the population.
f_2	Take the average of the crowding distances of all individuals.
f_3	The current optimization progress, represented as $1 - i/T$, where i denotes the current optimization step.
f_4	Average improvement rate of the current population compared to the previous generation.
f_5	Average standard deviation of objective function values.
f_6	The operators currently in use.
f_7	The best solution in the population.

and after optimization (with a "No new non-dominated individuals found" annotation when applicable), and 2) the evolutionary operators employed during that iteration. For the optimization features, Yi et al. (2022) categorized them into three types: search-dependent, solution-dependent, and instance-dependent features. We select the features outlined in Table 2, where $f_1 - f_6$ represent search-dependent characteristics for monitoring the optimization process, while f_7 and f_8 denote solution-dependent characteristics associated with solution encoding. Notably, instance-dependent features are excluded from our framework as they typically depend on problem-specific properties, which would compromise the generalizability of our proposed framework.

4. EXPERIMENTS

4.1 Experimental Settings

In our experimental setup, each unmanned aerial vehicle (UAV) is equipped with three base sensors denoted as s_1 , s_2 , s_3 , along with four composite configurations: $\{s_1, s_2\}$,

$\{s_1, s_3\}$, $\{s_2, s_3\}$, and $\{s_1, s_2, s_3\}$. These sensors are allocated to 12 targets, with cost parameters and collaborative coefficients specified in Table 3. Target priority coefficients and information gain settings remain consistent with the GPSO study (Yang et al., 2011).

Table 3. The parameters of sensors.

Sensors	Cost	Cooperation
s_1	1.000	1.000
s_2	1.500	1.000
s_3	2.000	1.000
$\{s_1, s_2\}$	3.000	1.526
$\{s_1, s_3\}$	3.500	1.607
$\{s_2, s_3\}$	4.000	2.109
$\{s_1, s_2, s_3\}$	5.500	2.713

For performance validation, LLM-MOEA is compared against common AOS strategies, including Random Strategy, Deep Double Q-Network (DDQN) (Yi et al., 2022), and LLM Step-Size Adaptation (LLM-SSA) (Custode et al., 2024). While the primary focus lies on parameter control, the methodology remains applicable to operator selection tasks. All methods share an identical operator pool containing three crossover operators—single-point, multi-point, and uniform crossover—and three mutation operators: swap, reverse, and shift mutation. For the two LLM-dependent approaches, LLM-MOEA and LLM-SSA, we consistently employ the open-source model llama3-70b-8192 for our experiments. In each experimental run, all algorithms are executed independently 20 times with a population size of 200 over 500 optimization iterations. Additionally, the operator selection mechanism is activated every 10 iterations, resulting in a total of 50 rounds of selection.

4.2 Overall performance

As demonstrated in Table 4, the proposed LLM-MOEA outperforms all three baseline methods across three critical metrics: Hypervolume (HV), Inverted Generational Distance (IGD), and Feasible Solution Ratio. Compared to the random policy, the DDQN-based approach achieves preliminary improvements in search efficiency (13.6% HV increase and a significant boost in feasible solutions), yet remains constrained by local convergence traps. In contrast, both LLM-driven methods exhibit superior performance, with LLM-MOEA delivering the most substantial gains: a 49.65% HV improvement over the random baseline, alongside a near-optimal IGD value (0.24 ± 0.38) and the highest feasible solution count.

Table 4. The experimental performance of different algorithms.

Method	HV	IGD	Feasible
Random	155.59 ± 35.29	3.30 ± 0.80	7.88 ± 1.82
DDQN	176.81 ± 25.43	2.61 ± 1.17	11.88 ± 2.47
LLM-SSA	205.26 ± 40.39	1.18 ± 0.85	11.88 ± 2.47
LLM-MOEA	232.84 ± 27.10	0.24 ± 0.38	14.58 ± 2.29

Fig. 3 further highlights the stability and consistency of LLM-MOEA, where its HV distributions across 20 independent trials show tighter clustering and higher median values compared to LLM-SSA. This observation validates

not only the method's enhanced convergence precision but also its robustness against initialization variance, positioning it as a promising approach for multi-objective constrained optimization.

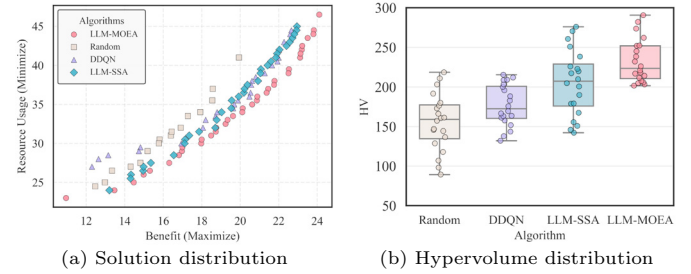


Fig. 3. Solution and hypervolume distributions of different algorithms in experiments.

4.3 Ablation Study

To verify the effectiveness of the two-component optimization logs in LLM-MOEA's prompts, we conducted an ablation study by introducing two variants: LLM-MOEA/T (removing optimization trajectories) and LLM-MOEA/F (removing state features). As illustrated in Fig. 4, both variants exhibit significant performance degradation compared to the full LLM-MOEA. These results confirm that both trajectory history and optimization features independently contribute critical guidance for selecting the best operators, thereby justifying the necessity of our prompt architecture.

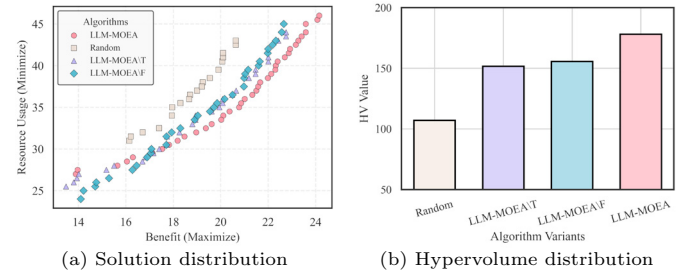


Fig. 4. Solution and hypervolume distributions of different variants in ablation study.

5. CONCLUSION

In this study, we propose LLM-MOEA, a novel operator selection framework that integrates LLMs with multi-objective optimization. The framework not only provides LLMs with optimization context but also incorporates domain-specific features tailored for Sensor-Target Assignment problems to enhance situational analysis. Experimental results verify that LLM-MOEA achieves operator selection decisions aligned with expert-level rationality while generating human-interpretable selection rationale. This work establishes a pioneering paradigm for combining LLMs with evolutionary optimization and highlights the untapped potential of LLMs in complex decision-making tasks. We plan to investigate dynamic feature selection mechanisms to adapt to evolving optimization states. Furthermore, efforts will focus on extending LLM-MOEA to broader real-world applications and rigorously evaluating its robustness under noisy and uncertain conditions.

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